SGT UNIVERSITY

FACULTY OF ENGINEERING AND TECHNOLOGY

LITERATURE REVIEW

B1 — SCORE AT SIX

First-Innings Score Prediction from Powerplay State

TRACK B • SEQUENCE AND UNCERTAINTY

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Academic Session 2026–27

1. <u>Introduction</u>

A twenty-over limited innings is played in Twenty20 (T20) cricket. Final team score depends on run rate and remaining wickets of the batting side. Prediction of the final score at the end of the innings has remained an ongoing issue in the analytics of cricket. It has been addressed in simple approaches by calculating run rate and in more sophisticated approaches with regression-based or machine-learning models using additional information such as the number of wickets lost and the stage of the innings. In this review, we consider a particular scenario in which final score prediction is performed after six overs in the innings. Six-over prediction is close to the end of the powerplay period and accounts for less than one-third of the innings. Since at such an early stage there is less information available, there is more uncertainty in predictions. One number may appear more reliable than the amount of information allows. Instead, there should be a number with an associated interval, which according to previous works, usually includes the actual score with the frequency of its coverage as stated.
The aim of this review is to provide an overview of current knowledge about prediction of a cricket score on the basis of incomplete information about the game, **modelling** techniques supported by previous works, confidence/uncertainty management in such problems, and an opening to predict six-over score with explicit uncertainty handling.

2. <u>Literature Review</u>

2.1 <u>Cricket and T20 Score Prediction</u>

Four pieces of research will be presented in the scope of this topic, addressing either the general concept of cricket score prediction during the game or machine learning approaches applied to T20 score prediction in particular.

Duckworth and Lewis [1] provide the basis of the topic and, despite being not related to the machine learning area at all, establish a concept of the innings scoring potential depending not only on the run rate, but on the number of overs and wickets remaining too. The method developed by the authors is aimed to solve problems of the Average Run Rate method, which failed in situations when wickets were not equally distributed between the sides. Even though the focus of this work is resetting of the target in case of interruptions rather than statistical analysis, it shows the presence of information carried by wickets about the innings scoring potential that run rate cannot account for.

Bailey and Clarke [2] use multiple linear regression on more than two thousand one-day international matches in order to predict the result during the game. They incorporate factors such as venue, form of the team and the current match state. They prove that their regression approach performs better than a simple run-rate projection.

Singh, Singla and Bhatia [3] build upon this concept, taking checkpoints every five overs during several innings. They demonstrate that linear regression model with the predictors of run rate, wickets fallen, venue, and batting team has lower error than a run-rate baseline at every checkpoint. At that, their win-prediction model is significantly less accurate at the beginning of the innings (68%) than at its end (91%). This proves that the predictive uncertainty is not consistent and is higher at the beginning of the innings, when there is a lack of match information.

Tonmoy et al. [4] use gradient-boosted trees (XGBoost) together with Lasso and Ridge regression to predict the first-innings score in T20 games based on ball-by-ball IPL data. They evaluate their models with mean absolute error, root mean squared error, and R^2. As follows from the results, both regularized linear models and gradient-boosted trees can be used for this score prediction task effectively, without any clear leader among them.

From this set of papers, there is a pattern that can be noticed. A simple run-rate projection represents a historical baseline, which is weak; the wickets fallen variable is the most useful predictor; and once the wickets variable is taken into account, both linear regression and tree-based models perform better than a run-rate projection. The limitations common for this set of works are: no prediction interval is reported in any of them; it is provided only as a point estimate or a classification accuracy, so there is no confidence estimation for the prediction. Another limitation is that the earliest part of the innings always has the highest error because of lack of state information, but none of the cricket-related papers here solves this problem.

2.2 In-Play Sports Prediction and Calibration

There are two other pieces of research on other sports that are not related to cricket and address a problem absent in any cricket paper: calibration of the probability estimation made alongside the prediction.

In Robberechts and Davis [5], a model for in-game estimation of probability of winning in soccer is designed with Bayesian interpretation of game-state features through eight seasons in eight leagues. The model is tested for calibration, not for accuracy, that is, if predicted probabilities match actual frequencies of outcomes. In Pelechrinis [6], a model for in-game estimation of probability of winning in the NFL game is tested in seven seasons of NFL play-by-play data for calibration and accuracy. The author finds out that a simple and understandable model performs as well as more complicated once calibration is focused, not just point accuracy.

Thus, both works show that prediction from incomplete game state only works when confidence is evaluated separately from accuracy. Both studies implicitly show that more complicated models are not necessarily better once calibrated performance is taken into account. Neither piece of research is about cricket or score regression but both of them establish a standard:

2.3 Machine Learning for Tabular Regression

There is another machine learning technique for tabular data that could be considered as an alternative to regression.

Gradient-boosted decision trees can be used for tabular data, which is exactly the kind of data needed for the score prediction problem. Chen and Guestrin [7] discuss XGBoost library and show that gradient boosting is a fast and accurate method for tabular data with mixed feature types and nonlinear interaction, which is the case for the match state variables (runs, wickets, and balls) since there is an interaction between boundaries in addition to their additive effect. In Tonmoy et al. [4], XGBoost along with some other regularised linear models is shown to perform well in predicting first-innings score in T20.

Thus, both pieces of research allow to consider gradient boosting and linear regression as good choices for this problem.

2.4 Prediction Uncertainty

The third category of research addresses the problem of predicting not single value but an interval along with checking confidence of the confidence level itself.

Meinshausen [8] extends random forest for the prediction of conditional quantiles, not mean values, and provides a non-parametric method for interpreting predictions of tree ensembles as low and high predictions—quantile regression forests. Romano, Patterson, and Candès [9] use the idea of combining quantile regression with conformal prediction and provide prediction intervals, which adapt to the context but keep the specified coverage in conformal prediction framework. They find out that the combination of quantile regression and conformal prediction can provide more informative prediction intervals than conformal prediction alone. Tyralis and Papacharalampous [10] survey several approaches for uncertainty estimation, which include quantile-based and conformal predictions, and note that the interval of uncertainty should be evaluated according to its coverage and width. A too wide interval may have high coverage but not be informative.

Thus, the literature shows that there are ways of introducing uncertainty estimation in regression prediction through quantile-based and conformal prediction. As for B1, it means that the model should not only predict the final score but also generate the corresponding interval, which can be checked for reliability on the new matches.

Table 1. Summary of reviewed literature.

#	Paper	Problem/Data	Method	Key Finding	What B1 Can Borrow
1	Duckworth & Lewis (1998)	Resetting targets in interrupted ODI matches	Deterministic resource table (overs remaining, wickets in hand)	Scoring potential depends jointly on overs and wickets, not run rate alone	Motivates including wickets alongside run rate rather than run rate alone

2	Bailey & Clarke (2006)	In-progress ODI outcome prediction; 2,200 ODI matches	Multiple linear regression	Regression outperforms a run-rate projection for in-progress prediction	Supports linear regression as a reasonable first model
3	Singh, Singla & Bhatia (2015)	Score/outcome prediction at 5-over checkpoints; ODI matches 2002–2014	Linear regression; Naive Bayes	Regression beats run rate at every checkpoint; accuracy lowest early in the innings	Suggests wider uncertainty should be expected at an early checkpoint
4	Tonmoy et al. (2024)	T20 first-innings score prediction; IPL ball-by-ball data	XGBoost, Lasso, Ridge regression	Gradient boosting and regularised regression both competitive for T20 score regression	Supports evaluating both linear regression and gradient boosting
5	Robberechts & Davis (2021)	In-game win probability; soccer, 8 seasons	Bayesian statistical model	Calibration is achievable and must be measured separately from accuracy	Supports evaluating calibration, not accuracy alone
6	Pelechrinis (2017)	In-game win probability; NFL, 7 seasons	Logistic-style model, simple vs. complex compared	A simple model matches complex ones once calibration is the criterion	Cautions against adding complexity without evidence
7	Chen & Guestrin (2016)	Scalable tree-boosting system; general tabular ML benchmarks	Gradient-boosted decision trees (XGBoost)	Gradient boosting is efficient and accurate for nonlinear tabular data	Supports gradient boosting as a candidate model
8	Meinshausen (2006)	Estimating the full conditional distribution; regression benchmarks	Quantile regression forests	Random forests generalise to give consistent quantile estimates	Provides a mechanism for producing low/high estimates
9	Romano, Patterson & Candès (2019)	Adaptive, valid-coverage prediction intervals; regression benchmarks	Conformalized quantile regression (CQR)	CQR gives shorter intervals than plain conformal prediction while keeping the coverage guarantee	Supports conformal methods for calibrated intervals
10	Tyralis & Papacharalampous (2024)	Review of predictive uncertainty estimation methods	Survey (quantile regression, QRF,	Coverage and interval width should be evaluated jointly	Supports reporting coverage and width together

3. Comparative Synthesis

Combining findings from the reviewed studies, there is evidence of several recurring themes. Run rate only projection as a baseline is outdated and poor-performing. Duckworth and Lewis came up with the resource approach to overcome the shortcomings of run rate only projection, and both Bailey and Clarke and Singh, Singla and Bhatia use run rate as a poor-performing baseline that is outperformed by regression models. Lost wickets **emerge** as an additional variable which is consistently related to more accurate prediction in cricket studies: it is included along with run rate or overs variable in all cricket-related studies here. Linear regression is demonstrated to be a good choice for the first step of the partial match prediction, while gradient-boosted trees provide great results when capturing non-linear dependencies between the match state variables without requiring a large training dataset. Moreover, more sophisticated model does not imply improved performance per se after basic point accuracy measures. More sophisticated model becomes useful only if it outperforms a simple model in terms of a certain metric.

The second recurring theme is what is not analysed in cricket score prediction studies. Cricket score prediction is concerned with the accuracy of point prediction only in terms of MAE, RMSE,

R^2, and classification accuracy. None of these cricket studies report prediction interval and test whether the predicted confidence level is correct. In the studies devoted to the in-play score prediction in other sports, the calibration of predictions is seen as a crucial element of the analysis, and both coverage and sharpness are used to assess the quality of the interval. This means that calibrated uncertainty estimation is a developed field, however, it has not been applied to cricket score prediction in the reviewed papers.

4. Research Gap

From the literature review above, it is possible to conclude that run rate projection is a poor baseline for predicting the final score in cricket innings and that incorporating lost wickets variable along with the regression tree model improves the accuracy of the predictions. It also indicates that gradient-boosted trees are a mature, extensively researched and well-grounded method for regression in tabular datasets with non-linear interaction of features, including cricket score prediction. Additionally, it proves that distribution-free prediction intervals with a guaranteed coverage in a finite sample through quantile regression forests, conformal prediction, or conformalized quantile regression techniques are extensively researched and can be used with any regression model.

What is not covered in the literature is a combination of the two above approaches for this particular task. None of the papers reviewed here provide calibrated prediction intervals with an empirical coverage estimate; all of them are focused on the point error or classification accuracy measure. None of the studies isolate a single checkpoint for prediction, such as the six-over mark in a twenty-over innings, as a standalone prediction task with its own prediction interval and coverage check. The closest study is Singh, Singla and Bhatia who perform multiple checkpoint analyses but do not consider an early checkpoint as an independent deliverable with the interval check. None of the papers performs this type of prediction on simulated data as opposed to the real data which would allow analyzing the structure of the data.

The gap in the literature is clear-cut: six-over T20 score prediction as a partial match state regression problem with the empirically tested prediction interval has not been studied in the reviewed papers.

5. Implications for B1

Table 2 connects the findings above to the modelling approach a six-over score-prediction task would reasonably adopt.

Table 2. Literature findings and their implications.

Literature finding	Implication for B1
Run rate alone is limited	CRR × 20 baseline
Wickets affect scoring potential	Include wickets_down
Regression works for partial-match prediction	Evaluate Linear Regression
Gradient boosting handles nonlinear tabular data	Evaluate Gradient Boosting

Point predictions do not quantify uncertainty	Produce prediction intervals
Conformal methods provide coverage guarantees	Investigate conformal prediction
Calibration matters	Report coverage and interval width

6. References

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